

FACULTY
OF MATHEMATICS
AND PHYSICS
Charles University

**BACHELOR THESIS**

Your Name Surname

# Thesis title

Name of the department

Supervisor of the bachelor thesis: Supername Supersurname
Study programme: study programme
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Prague YEAR

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Dedication. It is nice to say thanks to supervisors, friends, family, book authors and food providers.

Title: Thesis title

Author: Your Name Surname

Department: Name of the department

Supervisor: Supername Supersurname, department

Abstract: Abstracts are an abstract form of art. Use the most precise, shortest sentences that state what problem the thesis addresses, how it is approached, pinpoint the exact result achieved, and describe the applications and significance of the results. Highlight anything novel that was discovered or improved by the thesis. Maximum length is 200 words, but try to fit into 120. Abstracts are often used for deciding if a reviewer will be suitable for the thesis; a well-written abstract thus increases the probability of getting a reviewer who will like the thesis.

Keywords: key words

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# Introduction

Introduction should answer the following questions, ideally in this order:

1. What is the nature of the problem the thesis is addressing?
2. What is the common approach for solving that problem now?
3. How this thesis approaches the problem?
4. What are the results? Did something improve?
5. What can the reader expect in the individual chapters of the thesis?

Expected length of the introduction is between 1–4 pages. Longer introductions may require sub-sectioning with appropriate headings — use `\section*` to avoid numbering (with section names like 'Motivation' and 'Related work'), but try to avoid lengthy discussion of anything specific. Any "real science" (definitions, theorems, methods, data) should go into other chapters.

It is very advisable to skim through a book about scientific English writing before starting the thesis. I can recommend '*Science research writing for non-native speakers of English*' by Glasman-Deal [1].

You may notice that this paragraph briefly shows different "types" of 'quotes' in TeX, and the usage difference between a hyphen (-), en-dash (–) and em-dash (—).

# Chapter 1

# Introduction to thesis writing

If you are reading this, you likely have to write a thesis to finish your Bachelor or Master studies. First, congratulations on getting this far! What is a thesis though?

A thesis is a monograph, a longer writing about a given single topic of choice. The topic is typically implied by the area of your studies — at the school of computer science, you will write about solving a problem with computers, at medical school you will write about an issue of health or anatomy, etc. To create the thesis, you typically proceed as follows:

1. You conduct a moderate review of current (or recent) *academic literature* on a specific topic,
2. with the acquired knowledge, you solve a narrowly specialized and *well-defined problem*,
3. through standard and *reproducible methods*, you determine the quality of your solution,
4. and finally, you express the entire story *comprehensibly*, clearly, and lucidly in a 20- to 60-page book, which becomes your thesis.

By submitting a thesis, student proves that they are capable of understanding the results of recent research, using them to scientifically guide and analyze their own work, and communicating the outcome comprehensibly to others.

The work on the thesis is best started by finding a supervisor and agreeing on the topic. You will typically need to write an annotation of the thesis, which comprises a condensed summary of the main idea and expected results. The annotation is submitted to your institute's administration, and after some time, you will be officially assigned a task of completing the thesis.

Before the thesis is assigned, you usually prepare yourself for the main work; e.g., by testing all the software libraries, tools, and methods that you will need,

exploring your target domain, environment and subjects, creating and testing various prototypes and proof-of-concepts, reading relevant academic literature, and refining your approach (and the thesis annotation) based on initial findings. The preparation phase is quite crucial, as it allows you to discover many potential roadblocks and refine your approach *before* committing to the full thesis. Generally, the work conducted in advance gives you much better chance to finish the thesis without surprises later. For similar reasons, one of the main tasks of the supervisor is to know the roadblocks that you do not see yet, and prevent you from committing to a thesis that is not finishable or defendable.

## 1.1 How to recognize a good thesis topic?

Before you officially register a thesis — and ideally before you start committing to the task at all — you should know the concrete instances of terms highlighted in the points 1–4 above. In particular, you should be able to clearly answer the following questions:

**Is my idea even relevant?** To get the answer, go to Scholar[1] and try to find articles that are not too old (say, at most 15–20 years, depending on the area), are not visibly outdated, and are relevant to your topic. If you cannot find any recent academic work on your topic, it might be too niche, already solved, or not academically interesting. Your supervisor will often help you find the latest research in the area.

**Is the problem I want to solve sufficiently "hard"?** No one expects you to produce a completely new and revolutionary solution to your problem. The method for solving your problem should be known and already explored — in particular, you are not expected to make any groundbreaking scientific discoveries! You should "just" properly use the existing methods in a slightly novel context of your problem. For example, you may apply a known approach in a way that no one has tried before, slightly modify the approach to enable use in a seemingly different domain, or combine existing approaches in a novel way. In other words: the problem should be non-trivial, but you should know in advance how to approach the solution.

**Is the problem well-defined?** Solve one problem, and solve it well. You will often encounter a thesis reviewer telling you that your problem is not really a problem — this is especially common for theses with topics such as "I'll make a video game about X". To provide an answer, you must have a direct,

[1] `https://scholar.google.com/`, or any other scientific literature search tool

undeniable proof that your problem is relevant and its solution constitutes at least a minor advancement in some area (ideally, the world will be much more beautiful and civilization will be better off with the problem solved). Ideally, you will be able to point out existing academic literature that states that your topic is indeed a problem worth solving.

**How will I explain the problem to others?** Write the main problem of your thesis down as a *hypothesis*, and design an *experiment* that you can conduct to either confirm or reject this hypothesis, or to clearly determine the boundaries of its validity. Some theses are not experiments per se, but may be easily reworded as a case study, user testing, performance analysis, exploratory work, or theoretical proof. Mainly, the hypothesis–experiment wording helps you to convey the difficulty of the task, and clearly point out the "unknowns" that you aim to uncover.

**Am I trying to solve an exceedingly hard problem?** Am I perhaps solving multiple problems at once? In such cases, you might be shooting yourself in the foot: You will lose a lot of time on something that is not necessary, and your thesis reviewer will get annoyed trying to spot the single problem that you are solving. If your thesis addresses multiple smaller problems, you must correctly frame them in a single, well-defined and topically consistent group of problems.

**How do I present my results?** In your thesis, you will have to prove that you actually solved something, and reviewers will, by default, actively doubt that you accomplished anything meaningful. Prepare a clear method for evaluating your results. In computer science, the most common evaluation methods include performance measurements (speed, memory usage, accuracy), comparative analysis (how does your solution compare to existing approaches?), case studies (real-world applications of your solution), user studies (how do people interact with and benefit from your work?), and theoretical analysis (including mathematical proofs and logical arguments). It is extremely useful to show plots, statistics, and various other visualized evidence.

**How do I interpret my results?** The main method of interpreting your output is by rigorously measuring its properties (see above) and showing that the measurements fit your goals. The measurements should provide undisputable supporting evidence to allow you claim the goals (and partial goals) that were (and were not) fulfilled. Ideally, you will also compare your results with ones from people who solved similar problems — besides measuring your own solution, you might also need to measure the

alternative implementations and approaches for this purpose, use a comparable measurement methodology (benchmark) to allow comparison to other benchmarked work, and ensure fair comparison conditions. *However, no one expects you to win this comparison! The thesis is not a competition!* A negative result is also a result, and thesis reviewers will understand it; especially if the apparent failure is well-supported by measurements and used as a groundwork for some constructive recommendation to prevent this kind of failure for others. Conversely, reviewers will show no understanding for irreproducible measurements, unfair comparisons, and hiding the negative results.

**How am I going to fill all these pages with text?** Can you write 20–60 moderately amusing pages about your work? If not, you will be able to gain inspiration from reading academic literature thoroughly and elaborating on your findings (see above), conducting your own experiments and analysis, and reflecting on the broader implications of your work. Typically, you can write a lot of good text simply by reviewing 2–3 research papers that your thesis builds upon, and re-framing them into the context of your problem.

**Am I doing it right?** To understand what is typically expected, look at some of the completed theses in your field. Look into thesis repositories from your institution,[2] examine what makes a thesis successful, and observe the different approaches to similar problems, writing styles and organizational structures. To get a good picture of the common issues and expectations, read the official reviews from supervisors and reviewers (these are typically deposited along with the thesis).

The additional questions below do not directly relate to topic choice, but you may still want to equip yourself with suitable answers rather early, as they naturally shape the execution of the whole thesis:

**Why do I have to explain everything in text again?** You may wonder why you need to "wrap" your existing results (programs, measurements, algorithms, methods) in additional paper, despite of the fact that they are already visibly correct and solve the problem. First, the human-readable thesis serves as a singular artifact ensuring that your results will be understood and interpretable through *centuries*![3] This starkly contrasts with other short-lived artifacts: Measurements will come out completely different on hardware that appears in 5 years from now, and computer code

[2] `https://dspace.cuni.cz/` for Charles University

[3] University archives allow you to read and understand a thesis from 18$^{\text{th}}$ century.

(as well as laboratory protocols) will typically cease to work in a decade because the ecosystem will develop. Second, the thesis serves as a comprehension check: Indirectly, to be able to teach your approach to your peers reliably, you need to understand it very well. Make your thesis such that it conveys a piece of your own knowledge further, towards future scholars.

**Am I finishing it in time?** Allocate a lot of time for writing, editing, and making good pictures. At the *extreme barest minimum*, allocate at least one month of work for fixing and incorporating the comments from your supervisor, especially if you never wrote a thesis before. Do not expect supervisors to read your input immediately, they generally do not have time for that — instead, reserve at least a week for each revision. Check the thesis submission requirements in advance. In particular, if you are supposed to print your thesis on paper, order the printing service with sufficient time reserve.

**Who will tell me what to do next?** You will. Remember that it is *your* thesis — while your supervisor (and others) will try to guide you, it is ultimately you who is responsible for your own work (and success). You are thus expected to take initiative; and many would argue that the main point of the theses is precisely to *show the initiative*. In particular, never wait for your supervisor to give you tasks. Be proactive, work independently, solve issues well before deadlines, workaround issues early to avoid getting stuck, come to meetings equipped with prepared questions and clearly summarized progress updates and milestones, and bring up your own ideas about next steps. Make sure that your topic motivates you into taking the initiative.

## 1.2 What to put into the first chapter?

First chapter usually builds the theoretical background necessary for readers to understand the rest of the thesis, including your own contribution. You should summarize and reference a lot of existing literature and research.

You should use the standard *citations*.

Use `\emph` command like this, to highlight the first occurrence of an important word or term. Reader will notice it, and hopefully remember the importance.

**Obtaining bibTeX citation** Go to Google Scholar[4], find the relevant literature, click the tiny double-quote button below the link, and copy the bibTeX entry.

**Saving the citation** Insert the bibTeX entry to the file `refs.bib`. On the first line of the entry you should see the short reference name — from Scholar,

This footnote is an acceptable way to 'cite' web pages or URLs. Documents without proper titles, authors and publishers generally do not form citations. For this reason, avoid citations of Wikipedia pages.

[4] `https://scholar.google.com`

it usually looks like `author2015title` — you will use that to refer to the citation.

**Using the citation** Use the `\cite` command to typeset the citation number correctly in the text; a long citation description will be automatically added to the bibliography at the end of the thesis. Always use a non-breakable space before the citing parenthesis to avoid unacceptable line breaks:

```
Trees utilize gravity to invade ye
noble sires~\cite{newton1666apple}.
```

**Why should I bother with citations at all?** For two main reasons:

- You do not have to explain everything in the thesis; instead you send the reader to refer to details in some other literature. Use citations to simplify the detailed explanations.
- If you describe something that already exists without using a citation, the reviewer may think that you *claim* to have invented it. Expectably, they will demand academic correctness, and, from your perspective, being accused of plagiarism is not a good starting point for a successful defense. Use citations to identify the people who invented the ideas that you build upon.

**How many citations should I use?** Cite any non-trivial building block or assumption that you use, if it is published in the literature. You do not have to cite trivia, such as the basic definitions taught in the introductory courses.

The rule of thumb is that you should read, understand and briefly review at least around 4 scientific papers. A thesis that contains less than 3 sound citations will spark doubt in reviewers.

There are several main commands for inserting citations, used as follows:

- Knuth [2] described a great system for typesetting theses.
- We are typesetting this thesis with LaTeX, which is based on TeX and METAFONT [2].
- TeX was expanded to LaTeX by Lamport [3], hence the name.
- Revered are the authors of these systems! [2, 3]

## 1.3 Some extra assorted hints before you start writing English

**Word order** Strictly adhere to the English word order rules. The sentences follow a fixed structure with a subject followed by a verb and an object (in this order). Exceptions to this rule must be handled specially, and usually separated by commas.

**Sentence structure** Do not write long sentences. One sentence should contain exactly one fact. Multiple facts should be grouped in a paragraph to communicate one coherent idea. Both the sentences and paragraphs should include various hints about their relation to the other ideas and paragraphs. These are typically materialized as adverbs or short sentence parts that clarify the cause–outcome and target–method–result relationship of the sentences in a paragraph. Such 'word glue' helps the readers to correctly draw the lines that hold their mental images of your thesis together, and ideally see the big picture of what you were trying to convey right from the first read.

Paragraphs are grouped in labeled sections for a sole purpose of making the navigation in the thesis easier. Do not use the headings as 'names for paragraphs' — the text should make perfect sense even if all headings are removed. If a section of your text contains one paragraph per heading, you might have wanted to write an explicit list instead.

Mind the rules for placing commas:

- Do not use the comma before subordinate clauses that begin with 'that' (like this one). English does not use subordinate clauses as often as Slavic languages because the lack of a suitable word inflection method makes them hard to understand. In scientific English, try to avoid them as much as possible. Ask doubtfully whether each 'which' and 'when' is necessary — most of these helper conjunctions can be removed by converting the clause to non-subordinate.

  As an usual example, *'The sentence, which I wrote, seemed ugly.'* is perfectly bad; slightly improved by *'The sentence that I wrote seemed ugly.'*, which can be easily reduced to *'The sentence I wrote seemed ugly.'*. A final version with added storytelling value could say *'I wrote a sentence but it seemed ugly.'*

- Use the *Oxford comma* before 'and' and 'or' at the end of a longer, comma-separated list of items. Certainly use it to disambiguate any possible mixtures of conjunctions: *'The car is available in red, red and green, and green versions.'* Remember that English 'or' is typically understood more like

'either this or that, but not both,' and the use of 'and' is much more appropriate in cases such as possibility overviews and example listings (like in this sentence).

- Consider placing extra commas around any parts of the sentence that break the usual word order, especially if they are longer than a single word.

**Nouns** Every noun needs a determiner ('a', 'the', 'my', 'some', ...); the exceptions to this rule, such as non-adjectivized names and indeterminate plural, are relatively scarce. Without a determiner, a noun can be easily mistaken for something completely different, such as an adjective or a verb.

Name all things with appropriate nouns to help both the reader and yourself, and do not hesitate to invent good names and labels for anything that you will refer to more than once. Proper naming will save you a lot of writing effort because you will not have to repeat descriptions such as *'the third output of the second benchmarked method of the improved set,'* instead you may introduce a labeling that will allow you to say just something like *'output $M2^{+}/3$'*. At the same time, this will reduce the risk that the reader will confuse the object with another one — for illustration, the long version of the previous example might very easily confuse with the second output of the third method. The same also applies to methods descriptions, algorithms, programs, testing datasets, theorems, use-cases, challenges and other things. As an example, *'the algorithm that organizes the potatoes into appropriate buckets'* shortens nicely as *'the potato bucketer'* and may be labeled as a procedure BucketPotatoes(), and *'the issue where the robot crashes into a wall and takes significant time to return to the previous task'* may be called just *'the crash–recovery lag'*.

**Verbs** Although English can express a whopping 65 base verb tenses and their variants, scientific literature often suppresses this complexity and uses only several basic tenses where the meaning is clearly defined. Typically, you state facts in present simple (*'Theorem 1 proves that Gadget B works as intended.'*), talk about previous work and experiments done in past simple (*'We constructed Gadget B from Gizmo C, which was previously prepared by Tinkerer et al.'*), and identify achieved results in present perfect (*'We have constructed Technology T.'*). Avoid using future tense, except for sections that explicitly describe future work — as a typical mistake, if you state that the thesis *will* describe something in later chapters, you imply that the description is not present there yet.

Do not write sentences in passive voice, unless you explicitly need to highlight that something has passively subjected itself to an action. Active voice is more preferable in the theses because it clearly highlights the actors and their contributions — typically, *'you did it'* instead of *'it was done'* by a mysterious

entity, which the reviewers rarely envision as yourself. Writing in active voice additionally benefits the explanation of complex processes: There, the word order forces you to identify the acting subject as the first word in the sentence, which further disambiguates how the individual process parts are triggered and ordered.

Try to avoid overusing gerunds (verbs that end with '-ing'). It is convenient to write shorter sentences by using gerunds as adjectives, but these are typically quite hard to understand because the readers may easily confuse the intended adjectives with verbs. If your sentence contains two gerunds close to each other, it may need a rewrite.

**Scientific writing resources** Consult the book by Glasman-Deal [1] for more useful details and recommended terminology for writing about the scientific research. Very pragmatically, the book by Sparling [4] describes many common mistakes that Czech and Slovak (and generally Slavic) writers make when writing English.

Although your goal is not to write an actual research paper, the bachelor and master theses are commonly viewed as a preparation for that goal. In turn, research-paper writing and presentation resources (such as one by Peyton Jones [5], together with the associated presentation[5]) also become quite helpful.

[5] `https://simon.peytonjones.org/great-research-paper/`

# Chapter 2

# More complicated chapter

After the reader gained sufficient knowledge to understand your problem in chapter 1, you can jump to your own advanced material and conclusions.

You will need definitions (see definition 1 below in section 2.1), theorems (theorem 1), general mathematics, algorithms (algorithm 1), and tables (table 2.1). Figures 2.1 and 3.1 show how to make a nice figure. See figure 2.2 for an example of TikZ-based diagram. Cross-referencing helps to keep the necessary parts of the narrative close — use references to the previous chapter with theory wherever it seems that the reader could have forgotten the required context. Conversely, it is useful to add a few references to theoretical chapters that point to the sections which use the developed theory, giving the reader easy access to motivating application examples.

See documentation of package booktabs for hints on typesetting tables. As a main rule, *never* draw a vertical line.

## 2.1 Example with some mathematics

**Definition 1** (Triplet)**.** *Given stuff $X$, $Y$ and $Z$, we will write a* triplet *of the stuff as $(X, Y, Z)$.*

**Theorem 1** (Car coloring)**.** *All cars have the same color. More specifically, for any set of cars $C$, we have*

$$(\forall c_1, c_2 \in C)\ \textsc{Colour}(c_1) = \textsc{Colour}(c_2).$$

*Proof.* Use induction on sets of cars $C$. The statement holds trivially for $|C| \leq 1$. For larger $C$, select 2 overlapping subsets of $C$ smaller than $|C|$ (thus same-colored). Overlapping cars need to have the same color as the cars outside the overlap, thus also the whole $C$ is same-colored. □

This is plain wrong though.

| Column A | Column 2 | Numbers | More |
|---|---|---|---|
| Asd | QWERTY | 123123 | – |
| Asd qsd 1sd | BAD | 234234234 | This line should be helpful. |
| Asd | INTERESTING | 123123123 | – |
| Asd qsd 1sd | PLAIN WEIRD | 234234234 | – |
| Asd | QWERTY | 123123 | – |
| Asd qsd 1sd | GOOD | 234234299 | – |
| Asd | NUMBER | **123123** | – |
| Asd qsd 1sd | DANGEROUS | 234234234 | (no data) |

**Table 2.1** An example table. Table caption should clearly explain how to interpret the data in the table. Use some visual guide, such as boldface or color coding, to highlight the most important results (e.g., comparison winners).

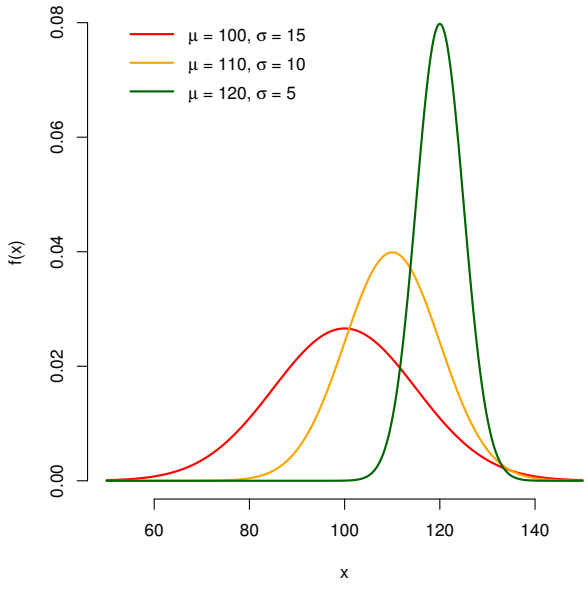



**Figure 2.1** A figure with a plot, not entirely related to anything. If you copy the figures from anywhere, always refer to the original author, ideally by citation (if possible). In particular, this picture — and many others, also a lot of surrounding code — was taken from the example bachelor thesis of MFF, originally created by Martin Mareš and others.

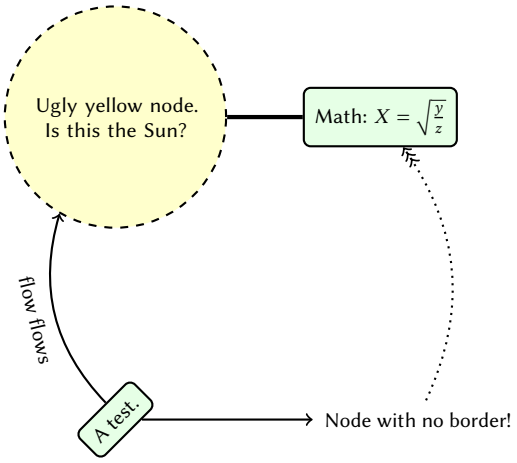



**Figure 2.2** An example diagram typeset with TikZ. It is a good idea to write diagram captions in a way that guides the reader through the diagram. Explicitly name the object where the diagram viewing should "start". Preferably, briefly summarize the connection to the parts of the text and other diagrams or figures. (In this case, would the tenative yellow Sun be described closer in some section of the thesis? Or, would there be a figure to detail the dotted pattern of the line?)

**Algorithm 1** Algorithm that executes an action with high probability. Do not care about formal semantics in the pseudocode — semicolons, types, correct function call parameters and similar nonsense from 'realistic' languages can be safely omitted. Instead make sure that the intuition behind (and perhaps some hints about its correctness or various corner cases) can be seen as easily as possible.

**function** ExecuteWithHighProbability($A$)
    $r \leftarrow$ a random number between 0 and 1
    $\varepsilon \leftarrow 0.00000000000000000000000000000000000042$
    **if** $r \geq \varepsilon$ **then**
        execute $A$ ..... *We discard the return value*
    **else**
        print: `Not today, sorry.`
    **end if**
**end function**

## 2.2 Extra typesetting hints

Do not overuse text formatting for highlighting various important parts of your sentences. If an idea cannot be communicated without formatting, the sentence probably needs rewriting anyway. Imagine the thesis being read aloud as a podcast — the storytellers are generally unable to speak in boldface font.

Most importantly, do not overuse bold text, which is designed to literally **shine from the page** to be the first thing that catches the eye of the reader. More precisely, use bold text only for 'navigation' elements that need to be seen and located first, such as headings, list item leads, and figure numbers.

Use underline only in dire necessity, such as in the previous paragraph where it was crucially important in ensuring that the reader remembers to never typeset boldface text manually again.

Use *emphasis* to highlight the first occurrences of important terms that the reader should notice. The feeling the emphasis produces is, roughly, "Oh my — what a nicely slanted word! Surely I expect it be important for the rest of the thesis!"

Finally, never draw a vertical line, not even in a table or around figures, ever. Vertical lines outside of the figures are ugly.

# Chapter 3

# Results and discussion

You should have a separate chapter for presenting your results (generated by the stuff described previously, in our case in chapter 2). Remember that your work needs to be validated rigorously, and no one will believe you if you just say that 'it worked well for you'.

Instead, try some of the following:

- State a hypothesis and prove it statistically
- Show plots with measurements that you did to prove your results (e.g. speedup). Use either `R` and `ggplot`, or Python with `matplotlib` to generate the plots.[1] Save them as PDF to avoid printing pixels (as in figure 3.1).
- Compare with other similar software/theses/authors/results, if possible
- Show example source code (e.g. for demonstrating how easily your results can be used)
- Include a 'toy problem' for demonstrating the basic functionality of your approach and detail all important properties and results on that
- Include clear pictures of 'inputs' and 'outputs' of all your algorithms, if applicable

It is sometimes convenient (even recommended by some journals, including Cell) to name the results sub-sections so that they state what exactly has been achieved. Examples follow.

[1] Honestly, the plots from `ggplot` look much better.

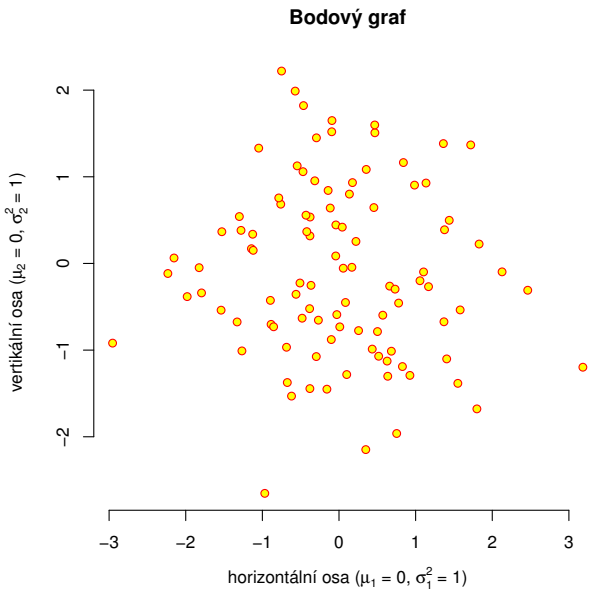



**Figure 3.1** This caption is a friendly reminder to never insert figures "in text," without a floating environment, unless explicitly needed for maintaining the text flow (e.g., the figure is small and developing with the text, like some of the centered equations, as in theorem 1). All figures *must* be referenced by number from the text (so that the readers can find them when they read the text) and properly captioned (so that the readers can interpret the figure even if they look at it before reading the text — reviewers love to do that).

## 3.1 SuperProgram is faster than OldAlgorithm

### 3.1.1 Scalability estimation

### 3.1.2 Precision of the results

## 3.2 Weird theorem is proven by induction

## 3.3 Amount of code reduced by CodeRedTool

### 3.3.1 Example

### 3.3.2 Performance on real codebases

## 3.4 NeuroticHelper improves neural network learning

## 3.5 Graphics and figure quality

No matter how great the text content of your thesis is, the pictures will always catch the attention first. This creates the very important first impression of the thesis contents and general quality. Crucially, that also decides whether the thesis is later read with joy, or carefully examined with suspicion.

Preparing your thesis in a way such that this first impression gets communicated smoothly and precisely helps both the reviewer and you: the reviewer will not have a hard time understanding what exactly you wanted to convey, and you will get a better grade.

Making the graphics 'work for you' involves doing some extra work that is often unexpected. At the same time, you will need to fit into graphics quality constraints and guidelines that are rarely understood before you actually see a bad example. As a rule of thumb, you should allocate at least the same amount of time and effort for making the figures look good as you would for writing, editing and correcting the same page area of paragraph text.

### 3.5.1 Visualize all important ideas

The set of figures in your thesis should be comprehensive and complete. For all important ideas, constructions, complicated setups and results there should be a visualization that the reader can refer to in case the text does not paint the 'mental image' sufficiently well. At the bare minimum, you should have at least 3

figures (roughly corresponding to the 3 chapters) that clearly and unambiguously show:

1. the context of the problem you are solving, optionally with e.g. question marks and exclamation marks placed to highlight the problems and research questions

2. the overall architecture of your solution (usually as a diagram with arrows, such as in figure 2.2, ideally with tiny toy examples of the inputs and outputs of each box),

3. the advancement or the distinctive property of your solution, usually in a benchmark plot, or as a clear demonstration and comparison of your results.

### 3.5.2 Make the figures comprehensible

The figures should be easily comprehensible. Surprisingly, that requires you to follow some common "standards" in figure design and processing. People are often used to a certain form of the visualizations, and (unless you have a very good reason) deviating from the standard is going to make the comprehension much more complicated. The common standards include the following:

- caption everything correctly, place the caption at an expectable position

- systematically label the plots with 'main' titles (usually in boldface, above the plot), plot axes, axis units and ticks, and legends

- lay out the diagrams systematically, ideally follow a structure of a bottom-up tree, a left-to-right pipeline, a top-down layered architecture, or a center-to-borders mindmap

- use colors that convey the required information correctly

  Although many people carry some intuition for color use, achieving a really correct utilization of colors is often very hard without previous experience in color science and typesetting. Always remember that everyone perceives color hues differently, therefore the best distinction between the colors is done by varying lightness of the graphics elements (i.e., separating the data by dark vs. light) rather than by using hues (i.e., forcing people to guess which one of salmon and olive colors means "better"). Almost 10% of the population have their vision impaired by some form of color vision deficiency, most frequently by deuteranomaly that prevents interpretation of even the most 'obvious' hue differences, such as green vs. red. Finally, printed colors look surprisingly different from the on-screen

colors. You can prevent much of these problems by using standardized palettes and well-tested color gradients, such as the ones from ColorBrewer[2] and ViridisLite[3]. Check if your pictures still look good if converted to greyscale, and use a color deficiency simulator to check how the colors are perceived with deuteranomaly.

Avoid large areas of over-saturated and dark colors:

- under no circumstances use dark backgrounds for any graphical elements, such as diagram boxes and tables — use very light, slightly desaturated colors instead
- avoid using figures that contain lots of dark color (as a common example, heatmaps rendered with the 'magma' color palette often look like huge black slabs that are visible even through the paper sheet, thus making a dark smudge on the neighboring page)
- increase the brightness of any photos to match the average brightness of the text around the figure

Remember to test your figures on other people — usually, just asking 'What do you think the figure should show?' can help you debug many mistakes in your graphics. If they think that the figure says something different than what you planned, then most likely it is your figure what is wrong, not the understanding of others.

Finally, there are many magnificent resources that help you arrange your graphics correctly. The two books by Tufte [6, 7] are arguably classics in the area. Additionally, you may find many interesting resources to help you with technical aspects of plotting, such as the `ggplot`-style 'Fundamentals' book by Wilke [8], and a wonderful manual for the TikZ/PGF graphics system by Tantau [9] that will help you draw high-quality diagrams (like the one in figure 2.2).

## 3.6 What is a discussion?

After you present the results and show that your contributions work, it is important to *interpret* them, showing what they mean in the wider context of the thesis topic, for the researchers who work in the area, and for the more general public, such as for the users.

Separate discussion sections are therefore common in life sciences where some ambiguity in result interpretation is common, and the carefully developed intuition about the wider context is sometimes the only thing that the authors

[2] `https://colorbrewer2.org`

[3] `https://sjmgarnier.github.io/viridisLite/`

have. Exact sciences and mathematicians do not need to use the discussion sections as often. Despite of that, it is nice to position your output into the previously existing environment, answering:

- What is the potential application of the result?
- Does the result solve a problem that other people encountered?
- Did the results point to any new (surprising) facts?
- How (and why) is the approach you chose different from what the others have done previously?
- Why is the result important for your future work (or work of anyone other)?
- Can the results be used to replace (and improve) anything that is used currently?

If you do not know the answers, you may want to ask the supervisor. Also, do not worry if the discussion section is half-empty or thoroughly pointless; you may remove it completely without much consequence. It is just a bachelor thesis, not a world-saving avenger thesis.

# Conclusion

In the conclusion, you should summarize what was achieved by the thesis. In a few paragraphs, try to answer the following:

- Was the problem stated in the introduction solved? (Ideally include a list of successfully achieved goals.)
- What is the quality of the result? Is the problem solved for good and the mankind does not need to ever think about it again, or just partially improved upon? (Is the incompleteness caused by overwhelming problem complexity that would be out of thesis scope, or any theoretical reasons, such as computational hardness?)

  This is quite common.
- Does the result have any practical applications that improve upon something realistic?
- Is there any good future development or research direction that could further improve the results of this thesis? (This is often summarized in a separate subsection called 'Future work'.)

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# Appendix A

# Using CoolThesisSoftware

Use this appendix to tell the readers (specifically the reviewer) how to use your software. A very reduced example follows; expand as necessary. Description of the program usage (e.g., how to process some example data) should be included as well.

To compile and run the software, you need dependencies XXX and YYY and a C compiler. On Debian-based Linux systems (such as Ubuntu), you may install these dependencies with APT:

```
apt-get install \
  libsuperdependency-dev \
  libanotherdependency-dev \
  build-essential
```

To unpack and compile the software, proceed as follows:

```
unzip coolsoft.zip
cd coolsoft
./configure
make
```

The program can be used as a C++ library, the simplest use is demonstrated in listing 1. A demonstration program that processes demonstration data is available in directory `demo/`, you can run the program on a demonstration dataset as follows:

```
cd demo/
./bin/cool_process_data data/demo1
```

After the program starts, control the data avenger with standard `WSAD` controls.

**Listing 1** Example program.

```
#include <CoolSoft.h>
#include <iostream>

int main() {
  int i;
  if(i = cool::ProcessAllData()) // returns 0 on error
    std::cout << i << std::endl;
  else
    std::cerr << "error!" << std::endl;
  return 0;
}
```